

## Topic 5: Waves and Particle Nature of Light

In order to develop their practical skills, students should be encouraged to carry out a range of practical experiments related to this topic. Possible experiments include determining the refractive index of solids and liquids, measuring the focal length of a lens, and using models of structures to investigate stress concentrations.

Mathematical skills that could be developed in this topic include using calculators to handle  $\sin x$ , identifying uncertainties in measurements and using simple techniques to determine uncertainty when data are combined.

This topic may be studied using applications that relate to waves and light, for example medical physics.

| Students should: |                                                                                                                                                                |
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| 59.              | understand the terms <i>amplitude, frequency, period, speed and wavelength</i>                                                                                 |
| 60.              | be able to use the wave equation $v = f\lambda$                                                                                                                |
| 61.              | be able to describe longitudinal waves in terms of pressure variation and the displacement of molecules                                                        |
| 62.              | be able to describe transverse waves                                                                                                                           |
| 63.              | be able to draw and interpret graphs representing transverse and longitudinal waves including standing/stationary waves                                        |
| 64.              | <b>CORE PRACTICAL 6: Determine the speed of sound in air using a 2-beam oscilloscope, signal generator, speaker and microphone.</b>                            |
| 65.              | know and understand what is meant by <i>wavefront, coherence, path difference, superposition, interference and phase</i>                                       |
| 66.              | be able to use the relationship between <i>phase difference</i> and <i>path difference</i>                                                                     |
| 67.              | know what is meant by a <i>standing/stationary</i> wave and understand how such a wave is formed, know how to identify nodes and antinodes                     |
| 68.              | be able to use the equation for the speed of a transverse wave on a string<br>$v = \sqrt{\frac{T}{\mu}}$                                                       |
| 69.              | <b>CORE PRACTICAL 7: Investigate the effects of length, tension and mass per unit length on the frequency of a vibrating string or wire.</b>                   |
| 70.              | be able to use the equation intensity of radiation $I = \frac{P}{A}$                                                                                           |
| 71.              | know and understand that at the interface between medium 1 and medium 2<br>$n_1 \sin \theta_1 = n_2 \sin \theta_2$ where refractive index is $n = \frac{c}{v}$ |
| 72.              | be able to calculate <i>critical angle</i> using $\sin C = \frac{1}{n}$                                                                                        |

**Students should:**

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| 75. | understand the term <i>focal length</i> of converging and diverging lenses                                                                                                                                              |
| 76. | be able to use ray diagrams to trace the path of light through a lens and locate the position of an image                                                                                                               |
| 77. | be able to use the equation power of a lens $P = \frac{1}{f}$                                                                                                                                                           |
| 78. | understand that for thin lenses in combination $P = P_1 + P_2 + P_3 + \dots$                                                                                                                                            |
| 79. | know and understand the terms <i>real image</i> and <i>virtual image</i>                                                                                                                                                |
| 80. | be able to use the equation $\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$ for a thin converging or diverging lens with the real is positive convention                                                                      |
| 81. | know and understand that magnification = image height/object height and $m = \frac{v}{u}$                                                                                                                               |
| 82. | understand what is meant by <i>plane polarisation</i>                                                                                                                                                                   |
| 83. | understand what is meant by <i>diffraction</i> and use Huygens' construction to explain what happens to a wave when it meets a slit or an obstacle                                                                      |
| 84. | be able to use $n\lambda = d\sin\theta$ for a diffraction grating                                                                                                                                                       |
| 85. | <b>CORE PRACTICAL 8: Determine the wavelength of light from a laser or other light source using a diffraction grating.</b>                                                                                              |
| 86. | understand how diffraction experiments provide evidence for the wave nature of electrons                                                                                                                                |
| 87. | be able to use the de Broglie equation $\lambda = \frac{h}{p}$                                                                                                                                                          |
| 88. | understand that waves can be transmitted and reflected at an interface between media                                                                                                                                    |
| 89. | understand how a pulse-echo technique can provide information about the position of an object and how the amount of information obtained may be limited by the wavelength of the radiation or by the duration of pulses |
| 90. | understand how the behaviour of electromagnetic radiation can be described in terms of a wave model and a photon model, and how these models developed over time                                                        |
| 91. | be able to use the equation $E = hf$ , that relates the photon energy to the wave frequency                                                                                                                             |
| 92. | understand that the absorption of a photon can result in the emission of a photoelectron                                                                                                                                |
| 93. | understand the terms <i>threshold frequency</i> and <i>work function</i> and be able to use                                                                                                                             |

**Students should:**

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| 95. | understand how the photoelectric effect provides evidence for the particle nature of electromagnetic radiation                                                                                                            |
| 96. | understand atomic line spectra in terms of transitions between discrete energy levels and understand how to calculate the frequency of radiation that could be emitted or absorbed in a transition between energy levels. |